

COURSE MANUAL "NON-LINEAR OPTIMISATION" (MAC 507)

MATHIAS STAUDIGL

1. COURSE DESCRIPTION

Optimisation methods are widely used in science, engineering and economics. Given this diverse set of users and application domains, it is essential for students and practitioners alike to develop an understanding of optimisation algorithms. Knowledge of the capabilities and limitations of these algorithms leads to better understanding of their impact, contributes to their explainability, and points the way to future research on improving and extending optimisation methods and software.

In this course, our goal is to give a rigorous introduction to the fascinating world of nonlinear optimisation, and some numerical solution methods that have proven useful in solving practical problems. Our focus in this course is on deterministic continuous optimisation methods. We do not enter the important field of stochastic optimization, but touch some topics in discrete optimization, mainly in connection with the extremely important model of a semi-definite program.

After mastering this course, students will know the standard optimality conditions used in general smooth optimization, knows what an SDP is, and why it is such a powerful tool, understands Lagrangian duality and is able to implement and analyze the standard NLP methods used in practice.

While not really necessary, it is a big advantage if you have already attended the course MAT404 Linear Programming.

2. TOPICS

In this course we cover the following topics

- Repetition of necessary facts from real calculus and linear algebra
- Optimality conditions for unconstrained and constrained smooth optimization problems (KKT conditions)
- Semi-Definite Programs and Interior Point Methods
- Gradient, Newton, Quasi-Newton Method and Trust-Region Methods
- Penalty and augmented Lagrangian methods
- SQP Methods

3. LITERATURE

There will be lecture notes available. The course follows closely the following excellent textbooks:

- A. Beck, Introduction to nonlinear optimization - Theory, Algorithms and Applications with MATLAB, MOS-SIAM Series on Optimization, 2014

- J. Nocedal & S.J. Wright. Numerical Optimization, 2nd Editions. Springer Series in Operations Research, 2000
- Carl Geiger, Christian Kanzow. Theorie und Numerik restringierter Optimierungsaufgaben, Springer-Verlag Berlin Heidelberg 2002

4. GRADING

This course consists of lectures and exercises classes. During the exercises you will present solutions to exercises. Some exercises are more of a theoretical nature, but in some others you will also be asked to implement a certain numerical method in Python. Active participation in the exercise classes is a necessary condition to be admitted to the final exam. This means concretely the following:

- (1) Submit Take-Home assignments within the indicated deadlines
- (2) Prepare tutorial exercises and make sure to actively participate in the tutorials.

At the end of the semester, there will be a written exam.